MUHAMMAD ILYAS RUSDI

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Surabaya, Jawa Timur

A highly communicative and sociable individual with a proven ability to build strong interpersonal relationships. My organizational experience has equipped me with the skills to collaborate effectively within teams, resolve conflicts, and reach consensus. Strong public speaking skills enable me to articulate ideas clearly and persuasively. adept at quickly adapting to dynamic environments

Education

MAN 2 Kota Malang - **Malang City, East Java.**

Senior High School, 91.50/100.00

- Ranked among the top 30 students eligible for SNBP 2024 from.
- Graduated as an accelerated student, completing the program in just two years
- top 20 national schools

Institut Teknologi Sepuluh Nopember - **Surabaya City, East Java.**

Student in Department of Informatics Engineering, Faculty of Electrical Engineering and Informatics Smart

Experiences Working and Projects

Real-Time UAV Video-to-Map Generation System Sep 2026 - Present

Research Assistant | Institut Teknologi Sepuluh Nopember (ITS)

- Engineered a robust video-to-map generation pipeline that processes UAV video streaming, applying advanced image stitching techniques to construct continuous and accurate aerial maps.
- Optimized algorithm stability and visual quality by mitigating scale drift via Hessian determinants, alongside implementing advanced seam finding and exposure compensation techniques.
- Validated the system's spatial accuracy and structural integrity by conducting rigorous quantitative evaluations utilizing reprojection error (RMSE), Structural Similarity Index (SSIM), and inlier ratios.

Immigration Coordinator for the Institut Teknologi Sepuluh Nopember Website - **Surabaya City, East Java** Jan 2026 - Aug 2026

- Led the end-to-end content and data migration of the university's website to a WordPress platform, effectively troubleshooting technical challenges to maintain an optimal site structure.
- Supervised a team of peers, delegating tasks and providing hands-on WordPress guidance to manage workflows and meet strict deployment deadlines.
- Conducted comprehensive quality assurance (QA) checks on the team's output to guarantee data integrity, responsive design, and correct formatting across all migrated pages.

DermaDiff - AI Skin Lesion Classification Sep 2025 - Feb 2026

AI Developer / Full-Stack AI Engineer

- https://dermadiff-web.vercel.app
- Engineered the core AI classification engine by implementing and fine-tuning models such as PanDerm, MedGemma 4B IT, and Stable Diffusion (via LoRA and DoRA) for optimized skin lesion analysis.
- Collaborated with a clinical expert to validate model accuracy against standard ABCD diagnostic rules and ensure high-fidelity performance on real-world smartphone camera images.
- Deployed the machine learning models into a fully functional, user-facing web application utilizing FastAPI for the backend and Next.js/React for the frontend, managing version control alongside a peer developer via GitHub.

AVALON AI - **Surabaya City, East Java** Nov 2024 - Dec 2025

Staff (machine learning engineer)

- Investigated deep learning training dynamics by analyzing the effects of data normalization on gradient flow and backpropagation stability in Convolutional Neural Networks (CNNs) to optimize model convergence.
- Conducted comprehensive evaluations of various activation functions across deep learning architectures, providing data-driven insights to improve neural network efficiency and predictive performance.
- Spearheaded DermaDiff as a flagship applied research initiative, bridging theoretical ML with medical diagnostics by fine-tuning advanced foundation models (PanDerm, MedGemma) and Stable Diffusion (LoRA/DoRA) for skin lesion classification.

Organisational Experience

Himpunan Mahasiswa Teknik Computer - Informatika ITS Jan 2026 - Present

Staff in Student Social Development

- Spearheaded internal student welfare initiatives, notably organizing a comprehensive Mental Health Seminar featuring professional psychologists to actively support the psychological well-being of Informatics students.
- Coordinated external community service and humanitarian outreach programs, mobilizing resources and student volunteers for disaster relief efforts and community educational empowerment campaigns.
- Bridged the gap between the university and the broader public by designing and executing targeted social impact programs that fostered empathy, social responsibility, and rapid crisis response.

ZENITRON - **Tuban, East java** - Present

Expert Staff (Human Resource Development)

the regional organization in Tuban which is a branch of Ini Lho ITS organization.

- provide lecture socialization for high school students from more than 3 schools in the Tuban area.become part of the operational staff in the "tryout sinergi" event which was attended by more than 300 participants.become an expert staff in the design of the 2025 breaking fast event in collaboration with external campuses such as ITB, UNIROW, and others
- Ensuring Program Alignment and Continuity. I play a crucial role in ensuring that each program under PSDM aligns with the vision and values of Forda Tuban. I help the team assess whether activities like training sessions, recruitment, or internal gathering.

SCHEMATICS 2026 - Present

Expert Staff (External Relation)

SCHEMATICS 2025 - **Indonesia** - Present

Staff (External Relation)

Schematics is an event created by Informatics Engineering students from the Sepuluh Nopember Institute of Technology which focuses on programming and logic competitions and introducing technology to the wider community with a current and interesting approach through attractive seminars and exhibitions.

- Contributed to expanding the event's network and visibility through strategic collaborations.
- Initiated and maintained communication with external partners, including companies, media, and communities.
- Strengthened skills in public speaking, proposal writing, and professional communication.

OKKBK 2025 Aug 2025

Staff (evaluator)

critical role within the Orientation Committee, responsible for the objective observation, assessment, and reporting of new student (participant) engagement, conduct, and performance.

- Evaluate and score participant assignments, reflection journals, or group projects according to a pre defined assessment rubric.
- Document any critical incidents, positive or negative, involving participant behavior.
- Evaluate and score participant assignments, reflection journals, or group projects according to a pre-defined assessment rubric.

Skills, Achievements & Other Experience

- **Soft Skills**: Leadership, Communication, Fast learner
- **Hard Skills** (2026): Python, SQL, C++, Java, JavaScript, PyTorch, TensorFlow, Matplotlib, Seaborn, FastAPI, React, Next.js, Tailwind CSS, Framer Motion, Git/GitHub, VS Code, Vercel, Google Colab, Nginx, GNS3,